DNA replication essentials

Complementary bases pair across antiparallel strands; new DNA extends 5' to 3'.

Complementary, antiparallel strands

Template 3' - T A C G G A - 5'

New DNA 5' - A T G C C T - 3'

New strand synthesis: 5' -> 3'

Protein / enzyme	Main job
Helicase	Separates parental DNA strands at the fork.
Primase	Makes primers that provide an extendable starting end.
DNA polymerase	Adds a matching nucleotide to an existing 3'-OH; synthesis is 5' -> 3'.
DNA ligase	Seals remaining backbone nicks after appropriate primer processing.

Leading versus lagging

Leading synthesis is continuous toward the fork. Lagging synthesis makes Okazaki fragments away from the fork. Both use the same 5' -> 3' synthesis rule because the templates are antiparallel.

Semiconservative outcome

Each daughter double helix has one parental strand and one new strand. Proofreading and repair reduce errors; an unrepaired mismatch can become a stable base-pair substitution after later replication.

DNA monomer: deoxyribonucleotide. Backbone linkage: phosphodiester bond. Complementary pairing: A-T and G-C. Enzyme details and primer processing differ between bacteria and eukaryotes.

DNA replication essentials

Self-check questions and explanations. Refer to the visual guide on the front.

What new DNA strand pairs with 3'-TACGGA-5'?

New strand: 5'-ATGCCT-3'. Its template is 3'-TACGGA-5'. A pairs with T and G with C; the two strands are antiparallel.

At which end does DNA polymerase add a nucleotide?

DNA polymerase adds to the existing 3'-OH, so the new strand extends 5' to 3'. In this diagram synthesis proceeds left to right along the displayed new strand.

What makes replication semiconservative?

Each daughter double helix has one parental strand and one new strand: semiconservative replication. An intact parental strand is retained, rather than half of each parental base being mixed into a new strand.

Why are leading and lagging synthesis different?

Opposite template orientations combined with only 5'-to-3' synthesis make leading synthesis continuous toward the fork and lagging synthesis discontinuous as Okazaki fragments away from the fork.

What do helicase, primase, polymerase and ligase each do?

Helicase separates strands; primase makes primers; DNA polymerase extends DNA using a template; ligase seals remaining backbone nicks after appropriate processing. Enzymes do not change the allowed synthesis direction.

Can an unrepaired mismatch become heritable, and must it change a protein?

An unrepaired mismatched base can act as a template, fixing a changed base pair in one lineage after replication. Some substitutions are silent or outside coding regions; others alter sequence, expression or function.